



Mangalore University

B.Sc. DEGREE

STATE EDUCATION POLICY  
(SEP-2024)

COURSE PATTERN AND SCHEME OF EXAMINATION

CORE SUBJECT: ZOOLOGY

Academic Year 2024 -2025 Onwards

## **PREAMBLE**

The syllabus of Core Zoology for the B.Sc. Degree Programme under the State Education Policy (SEP–2024) has been framed in accordance with the curriculum framework and academic guidelines issued by the University. The revised syllabus is introduced with effect from the academic year 2024–2025 onwards.

The syllabus is designed to provide students with a comprehensive understanding of various branches of Zoology and to meet the evolving needs of higher education. It includes topics of contemporary relevance such as biodiversity conservation, ecology, genetics, biotechnology, animal behaviour, environmental biology, and applied Zoology.

The curriculum emphasizes conceptual understanding, practical skills, scientific temper, and research aptitude through an integrated approach to teaching and learning.

### **The highlights of the syllabus are as follows:**

- Greater emphasis is given to the study of local fauna, biodiversity, ecology, and conservation.
- An integrated and multidisciplinary approach has been adopted to correlate various subdivisions of Zoology.
- The syllabus includes emerging areas of biological sciences along with provisions for innovative and experiential learning.
- The syllabus of each paper is unitized, and the question paper pattern is structured to ensure balanced coverage within each unit.
- Practical components and field-based learning are incorporated to strengthen observational and analytical skills among students.
- The Regulations and Scheme of Examination are prepared as per the guidelines issued by the University.

### **U.G. Board of Studies in Zoology**

Dr. Mohammed Mustak B. – Chairperson

Dr. Vijayakumar K. M. – Member

Dr. Nagarathna K. A. – Member

Mrs. Sushama C. – Member

Mrs. Manasa Shetty – Member

**MANGALORE UNIVERSITY**  
**B.Sc. DEGREE**  
**STATE EDUCATION POLICY (SEP-2024)**  
**COURSE PATTERN AND SCHEME OF EXAMINATION**  
**CORE SUBJECT: ZOOLOGY**

| Core/<br>Elective                                            | CourseCode              | Title of thecourse                                  | Instructi<br>on<br>(Hrs) | Duration of<br>the<br>Examination<br>(Hrs) | Max. Marks |    |       | Credits |
|--------------------------------------------------------------|-------------------------|-----------------------------------------------------|--------------------------|--------------------------------------------|------------|----|-------|---------|
|                                                              |                         |                                                     |                          |                                            | Exam       | IA | Total |         |
| I Semester B.Sc.                                             |                         |                                                     |                          |                                            |            |    |       |         |
| Group I<br>Core<br>Subject                                   | Theory<br>BSCZOCS101    | Animal Diversity-I<br>(Non -Chordata)               | 4                        | 3                                          | 80         | 20 | 100   | 3       |
|                                                              | Practical<br>BSCZOPS101 | Animal Diversity-I<br>(Non -Chordata)               | 4                        | 3                                          | 40         | 10 | 50    | 2       |
| Total number of Credits for Core Subject in I Semester: 05   |                         |                                                     |                          |                                            |            |    |       |         |
| II Semester B.Sc.                                            |                         |                                                     |                          |                                            |            |    |       |         |
| Group I<br>Core<br>Subject                                   | Theory<br>BSCZOCS201    | Animal Diversity<br>- II (Chordata)                 | 4                        | 3                                          | 80         | 20 | 100   | 3       |
|                                                              | Practical<br>BSCZOPS201 | Animal Diversity<br>- II (Chordata)                 | 4                        | 3                                          | 40         | 10 | 50    | 2       |
| Total number of Credits for Core Subject in II Semester: 05  |                         |                                                     |                          |                                            |            |    |       |         |
| III Semester B.Sc.                                           |                         |                                                     |                          |                                            |            |    |       |         |
| Group I<br>Core<br>Subject                                   | Theory<br>BSCZOCS301    | Physiology,<br>Biochemistry and<br>Immunology       | 4                        | 3                                          | 80         | 20 | 100   | 3       |
|                                                              | Practical<br>BSCZOPS301 | Physiology,<br>Biochemistry and<br>Immunology       | 4                        | 3                                          | 40         | 10 | 50    | 2       |
| Group II<br>Elective/<br>Optional                            | BSCZOES301              | Parasitology and<br>Vector Biology                  | 2                        | 2                                          | 40         | 10 | 50    | 2       |
| Total number of Credits for Core Subject in III Semester: 07 |                         |                                                     |                          |                                            |            |    |       |         |
| IV Semester B.Sc.                                            |                         |                                                     |                          |                                            |            |    |       |         |
| Group I<br>Core<br>Subject                                   | Theory<br>BSCZOCS401    | Histology,Animal<br>Behavior and<br>Applied Zoology | 4                        | 3                                          | 80         | 20 | 100   | 3       |

|                                                                              |                         |                                                                                                                          |   |   |    |    |     |   |
|------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------|---|---|----|----|-----|---|
|                                                                              | Practical<br>BSCZOPS401 | Histology, Animal<br>Behavior and<br>Applied Zoology                                                                     | 4 | 3 | 40 | 10 | 50  | 2 |
| Group II<br>Elective/<br>Optional                                            | BSCZOES401              | Aquarium Fish<br>Keeping                                                                                                 | 2 | 2 | 40 | 10 | 50  | 2 |
| Total number of Credits for Core Subject in IV Semester: 07                  |                         |                                                                                                                          |   |   |    |    |     |   |
| <b>V Semester B.Sc.</b>                                                      |                         |                                                                                                                          |   |   |    |    |     |   |
| Group I<br>Core<br>Subject                                                   | Theory<br>BSCZOCS501    | Cell Biology and<br>Biotechnology                                                                                        | 3 | 3 | 80 | 20 | 100 | 3 |
|                                                                              | Theory<br>BSCZOCS502    | Genetics,<br>Biostatistics,<br>Evolution and<br>Paleontology                                                             | 3 | 3 | 80 | 20 | 100 | 3 |
|                                                                              | Practical<br>BSCZOPS501 | Cell Biology,<br>Biotechnology,<br>Genetics,<br>Biostatistics,<br>Evolution and<br>Paleontology                          | 4 | 3 | 40 | 10 | 50  | 2 |
| Total number of Credits for Core Subject in V Semester: 08                   |                         |                                                                                                                          |   |   |    |    |     |   |
| <b>VI Semester B.Sc.</b>                                                     |                         |                                                                                                                          |   |   |    |    |     |   |
| Group I<br>Core<br>Subject                                                   | Theory<br>BSCZOCS601    | Reproductive<br>Biology and<br>Developmental<br>Biology                                                                  | 3 | 3 | 80 | 20 | 100 | 3 |
|                                                                              | Theory<br>BSCZOCS602    | Environmental<br>Biology,<br>Toxicology and<br>Wildlife Biology                                                          | 3 | 3 | 80 | 20 | 100 | 3 |
|                                                                              | Practical<br>BSCZOPS601 | Reproductive<br>Biology,<br>Developmental<br>Biology,<br>Environmental<br>Biology,<br>Toxicology and<br>Wildlife Biology | 4 | 3 | 40 | 10 | 50  | 2 |
| Total number of Credits for Core Subject in VI Semester: 08                  |                         |                                                                                                                          |   |   |    |    |     |   |
| Group II<br>Compulsory<br>course                                             | BSCZOPRS<br>401/501/601 | *Skill enhancement<br>programme -<br>Project work / Field<br>training                                                    | 2 | 3 | 40 | 10 | 50  | 2 |
| Total number of Credits for Core Subject in I-VI Semesters: 42               |                         |                                                                                                                          |   |   |    |    |     |   |
| * Compulsory course to be offered in either IV/V/VI Semester by the college. |                         |                                                                                                                          |   |   |    |    |     |   |

**MANGALORE UNIVERSITY**  
**STATE EDUCATION POLICY (SEP-2024)**  
**B.Sc. DEGREE - ZOOLOGY**

|                                       |                                                       |                                       |               |
|---------------------------------------|-------------------------------------------------------|---------------------------------------|---------------|
| <b>Program Name</b>                   | <b>B.Sc.</b>                                          | <b>SEMESTER</b>                       | <b>I</b>      |
| <b>Course Title</b>                   | <b>ANIMAL DIVERSITY-I (NON-CHORDATA)<br/>(THEORY)</b> |                                       |               |
| <b>Course Code</b>                    | <b>BSCZOCS101</b>                                     | <b>No. of Credits</b>                 | <b>3</b>      |
| <b>Contact hours</b>                  | <b>48 Hrs<br/>(4 hours/week)</b>                      | <b>Duration of SEA/Exam</b>           | <b>3 Hrs.</b> |
| <b>Formative Assessment<br/>Marks</b> | <b>20</b>                                             | <b>Summative Assessment<br/>Marks</b> | <b>80</b>     |

**COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Explain the principles of animal classification, biodiversity concepts, and phylogenetic relationships among non-chordates.

**CO2:** Describe the general characters, classification, morphology, and life cycles of important protozoans, poriferans, cnidarians, and ctenophores.

**CO3:** Analyze the structural organization, adaptations, and pathogenicity of platyhelminthes, nematodes, and annelids.

**CO4:** Interpret the evolutionary significance and distinguishing features of arthropods, molluscs, echinoderms, and related groups.

**Unit - I: Introduction, Biodiversity, Protozoa**

**1.1 Introduction**

**2 Hrs**

Principles of animal classification - Binomial nomenclature, Linnaean hierarchy; Criteria for animal classification - body layers, coelom, body symmetry, metamerism, cephalization; Definition of species; Phylogeny; Classification of Animal Kingdom up to phylum.

**1.2 Biodiversity**

**4 Hrs**

Levels of biodiversity - species, genetic and ecosystem level diversity; Concept of Biodiversity hotspots; Biodiversity hotspots of India with emphasis on Western Ghats & Himalaya hotspots.

**1.3 Phylum: Protozoa**

**6 Hrs**

General characters of the phylum and classification up to classes with suitable examples; Structure and life history of malarial parasite (*Plasmodium vivax*) and human parasitic protozoan (*Entamoeba histolytica*).

**Unit – II: Porifera, Coelenterata, Ctenophora**

**2.1 Phylum: Porifera**

**6 Hrs**

General characters of the phylum and classification up to classes with suitable examples; Sycon-morphology; canal system in sponges.

## **2.2 Phylum: Cnidaria and Ctenophora**

**6 Hrs**

General characters of the phylum and classification up to classes with suitable examples; Polymorphism in *Physalia* and *Halistemma*; Coral reefs – Fringing, Barrier, Atoll; Structure of Corallite; Metagenesis in *Obelia*.

Distinctive characters of Ctenophora, Externals of *Pleurobrachia*.

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## **Unit - III: Platyhelminthes, Nemathelminthes and Annelida**

### **3.1 Phylum: Platyhelminthes**

**4 Hrs**

General characters of the phylum and classification up to classes with suitable examples; Structure and life history of liver fluke and tapeworm.

### **3.2 Phylum: Nemathelminthes**

**4 Hrs**

General characters of the phylum with suitable examples; External characters, life cycle and pathogenicity and preventive measures of *Ascaris*.

### **3.3 Phylum: Annelida**

**4 Hrs**

General characters of the phylum and classification up to classes with suitable examples; Tubicolous adaptations in *Sabella* and *Chaetopterus*; External features and life history of earthworm.

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## **Unit – IV: Arthropoda, Onychophora, Mollusca and Echinodermata**

### **4.1 Phylum: Arthropoda and Onychophora**

**5Hrs**

General characters of the phylum and classification up to classes with suitable examples; Externals of marine prawn (*Penaeus*) with detailed account of appendages; Metamorphosis in insects – Definition, types with examples – ametabola, hemimetabola and holometabola; Externals of *Peripatus*.

### **4.2 Phylum: Mollusca**

**3Hrs**

General characters of the phylum and classification up to classes with suitable examples; External features of *Unio* and *Pila*.

### **4.3 Phylum: Echinodermata**

**4Hrs**

General characters of the phylum and classification up to classes with suitable examples; External features and water-vascular system of *Asterias*; Larval forms of Echinodermata – Bipinnaria, Echinopluteus.

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### **Note:**

- Local examples with common and scientific names are to be given more emphasis for all the groups.
  - While selecting the examples, only such of the salient features of the examples have to be mentioned which are necessary to explain the general characters of the phylum/class.
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**REFERENCES:**

1. Adam, S. 1990. A Students Text Book of Zoology, Vol. I, II & Vol. III. Low Price Publications, New Delhi.
2. Agarwal, V.K. 2017. Zoology for Degree Students: Non-Chordata, S. Chand & Company, New Delhi.
3. Ayyar, E. 1982. A Manual of Zoology Vol. I, Part I & II, S. Vishwanathan Pvt. Ltd.
4. Barnes, R.S.K., Calow. P., Olive, P.J.W., Golding, D.W., Spicer, J.I. 2002. The Invertebrates: Synthesis, Blackwell Publishing.
5. Dhami, P.S. & Dhami, J.K. 2021. Invertebrate Zoology, 5<sup>th</sup> Edition, R. Chand & Co. New Delhi.
6. Hickman, C., Roberts, L.S., Keen, S.L., Larson, A. and Eisenhour, D. 2018. Animal Diversity, McGraw-Hill.
7. Holland, P. 2011. The Animal Kingdom: A Very Short Introduction, Oxford University Press.
8. Hussain, S. A. & Achar, K.P. 1999. Biodiversity of the Western Ghats Complex of Karnataka. Resource Potential and Sustainable Utilization. Mangalore: Biodiversity Initiative Trust.
9. Jordan, E.L. & Verma, P.S. 2022. Invertebrate Zoology, S. Chand & Company, New Delhi.
10. Kotpal, R.L. 2017. Modern Text Book of Invertebrates, Rastogi Publications, Meerut.
11. Kotpal, R.L. 2017. Protozoa to Echinodermata (Phylum Series), Rastogi Publications, Meerut.
12. Lal, S.S. 2016. A Text book of Practical Zoology – Invertebrates, Rastogi Publications.
13. Verma, P.S. 2013. A Manual of Practical Zoology – Invertebrates, S. Chand & Co. New Delhi.

|                                   |                                                      |                                   |              |
|-----------------------------------|------------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                         | <b>SEMESTER</b>                   | <b>I</b>     |
| <b>Course Title</b>               | <b>ANIMAL DIVERSITY-I (NON-CHORDATA) (PRACTICAL)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOPS101</b>                                    | <b>No. of Credits</b>             | <b>2</b>     |
| <b>Contact hours</b>              | <b>4 Hrs/week</b>                                    | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>10</b>                                            | <b>Summative Assessment Marks</b> | <b>40</b>    |

### COURSE OUTCOME:

**After completion of the course, students will be able to:**

**CO1:** Identify and classify important non-chordate specimens, slides, and larval forms.

**CO2:** Demonstrate knowledge of dissection techniques and organ systems of selected invertebrates.

**CO3:** Prepare temporary and whole mounts of zoological materials using standard laboratory procedures.

**CO4:** Develop observational, diagrammatic, and practical skills in non-chordate zoology.

### Museum specimens and slides.

- Commonly available specimens cited in the list of examples are to be selected for practical's.
- Dissections (Demonstration only)
  - Prawn: Nervous system
  - Earthworm: Nervous system
  - Leech: a) Digestive system  
b) Reproductive system
  - Cockroach: a) Digestive system  
b) Nervous system
  - Male and female reproductive systems
- Observations of hay infusion culture to study living protozoans like *Euglena*, *Paramecium*, *Vorticella*, *Amoeba* etc.
- Mounting and Whole mount preparations
- Demonstration of mounting of the following:
  - Prawn: Appendages
  - Leech: Salivary gland cells and Jaw
  - Cockroach: Salivary glands and Mouth parts
  - Earthworm: Ovary and Body setae
- Whole mount preparation: Cnidarian colonies - *Obelia*, *Sertularia*, *Pennaria*, *Tubularia* (any two); Crustacean larvae – Nauplius, Zoea, Mysis (any two) - Processing and mounting.

### LIST OF MUSEUM SPECIMENS AND SLIDES

- Slides of *Elphidium*, *Euglena*, *Plasmodium*, *Paramecium*, and *Vorticella*.
- Specimens of *Euplectella*, *Sycon*, slides of sponge spicules.
- Obelia*, *Physalia*, *Aurelia*, *Sea anemone*, *Fungia*.



4. Planaria, Liver fluke, Tapeworm, *Ascaris* (Male and Female).
  5. *Nereis*, *Chaetopterus*, *Pheretima*, *Leech*.
  6. *Carcinus* (male or female crab), *Peripatus*, *Lepas*, *Scolopendra*, *Limulus* and *Palamnaeus* (Scorpion).
  7. *Chiton*, *Dentalium*, *Xancus*, *Pila*, *Unio*, *Nautilus*, *Octopus*.
  8. *Asterias* (Star fish), *Ophiothrix* (Brittle star), *Echinus* (Sea-urchin), *Cucumaria* (Sea-Cucumber), *Antedon* (Sea-lily), Bipinnaria larva, Pluteus larva.
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**SCHEME OF PRACTICAL EXAMINATION**  
**B.Sc. ZOOLOGY: I SEMESTER**  
**Course Title: ANIMAL DIVERSITY- I (NON-CHORDATA)**  
**Code: BSCZOPS101**

**Duration: 3 hours**

**Max. Marks: 40**

- 
- |      |                                                                                                                                             |                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| I.   | Dissection - Identify, draw labeled diagram and comment on the systems A and B.                                                             |                   |
|      |                                                                                                                                             | <b>2 x 4 = 08</b> |
|      | (Identification of the system - 1 Mark; Labelled diagram of the entire system - 2 Marks; Description - 1 Mark) (A & B in different animals) |                   |
| II.  | Mounting – Identify and comment on C. [Any one item from C (i)]                                                                             | <b>03</b>         |
|      | (Identification -1 Mark; Diagram -1 Mark; Minimum two unique characters -1 Mark)                                                            |                   |
| III. | Mounting – Make a stained, temporary mounting of the given material D.                                                                      | <b>03</b>         |
|      | [Any one item from C (ii)] (Stained preparation-2 Marks; Procedure-1 Mark)                                                                  |                   |
| IV.  | Identify, classify, draw labeled diagram and comment on E, F, G, and H.                                                                     |                   |
|      | (1 slide, 3 specimens). <b>4x4=16</b>                                                                                                       |                   |
|      | (Identification - ½ Mark; Classification -½ Mark; Labelled diagram -1 Mark; Four Comments - 2 Marks)                                        |                   |
| V.   | Class records                                                                                                                               | <b>10</b>         |
|      |                                                                                                                                             | <b>TOTAL = 40</b> |
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**Note:**

- Questions must be framed as per the scheme provided.
- Internal assessment marks to be allotted after conducting one practical test at the end of the semester.

|                                   |                                                |                                   |              |
|-----------------------------------|------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                   | <b>SEMESTER</b>                   | <b>II</b>    |
| <b>Course Title</b>               | <b>ANIMAL DIVERSITY-II (CHORDATA) (THEORY)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOCS201</b>                              | <b>No. of Credits</b>             | <b>3</b>     |
| <b>Contact hours</b>              | <b>48 Hrs<br/>(4 hours/week)</b>               | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative assessment marks</b> | <b>20</b>                                      | <b>Summative assessment marks</b> | <b>80</b>    |

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### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Explain the general characteristics and classification of chordates and vertebrates.

**CO2:** Describe the morphology, adaptations, and evolutionary significance of protochordates, cyclostomes, fishes, and amphibians.

**CO3:** Analyze the structural features, poison apparatus, and adaptations in reptiles and birds.

**CO4:** Interpret the classification, organ systems, and exoskeletal structures of mammals.

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### **Unit - I: Hemichordata, Chordata, and Cyclostomata**

#### **1.1 Hemichordata**

**3 Hrs**

General characters of the phylum and external features of *Balanoglossus*; Tornaria larva.

#### **1.2 Chordata**

**1 Hrs**

General characters of Chordata and outline classification up to subphyla.

#### **1.3 Protochordata**

**3 Hrs**

Characters of Urochordata and Cephalochordata with examples; External features of *Herdmania* and *Branchiostoma*.

#### **1.3. Vertebrata**

**1 Hrs**

General characters of Vertebrata; Outline classification up to classes.

#### **1.4 Cyclostomata**

**4 Hrs**

General characters; External features and differences between Lamprey (*Petromyzon*) and Hag fish (*Myxine*), Structure of Ammocoetes larva and its metamorphosis.

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### **Unit - II: Pisces and Amphibia**

**2.1. Pisces** - General characteristics and aquatic adaptations of fishes.

**2 Hrs**

#### **2.2 Chondrichthyes and Osteichthyes**

**3 Hrs**

General characters of Chondrichthyes with examples; General characters of Osteichthyes with examples.

## **2.3 Amphibia**

**2 Hrs**

General characters and classification up to orders; Distinguishing features of Anura, Apoda and Urodela with suitable examples.

### **2.3.1 Endoskeleton of Frog**

**5 Hrs**

Skull, lower jaw, hyoid apparatus, vertebral column, pectoral and pelvic girdles, limb skeleton.

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## **Unit - III: Reptilia and Aves**

### **3.1 Reptilia**

**6 Hrs**

General characters and classification up to orders (living orders only) with suitable examples; Temporal fossae and arcades in reptiles; Indian snakes – Examples of poisonous and Non-poisonous snakes; Distinguishing poisonous from non-poisonous snakes; Poison apparatus and its working mechanism; Snake venom and anti-venom.

### **3.2 Aves**

**6 Hrs**

General characters and classification; Distinctive features of Archaeornithes and Neornithes with reference to Palaeognathae, Impennae and Neognathae giving suitable examples; Flight adaptations in birds.

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## **Unit - IV: Mammalia**

### **4.1 Classification and distinctive features**

**4Hrs**

General characters and classification up to subclasses; Distinctive features of Prototheria, Metatheria and Eutheria with important examples.

### **4.2. Important characters of following Eutherian orders with example**

**5Hrs**

Primates, Chiroptera, Cetacea, Perissodactyla, Artiodactyla, Carnivora, Rodentia and Proboscidea.

### **4.3. Organ systems**

**2Hrs**

Study of digestive system and reproductive system of rat.

### **4.4 Exoskeletal structures**

**1Hrs**

Structure of horns, antlers, hooves & hairs

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#### **Note:**

- Local examples with common and scientific names are to be given more emphasis for all the groups.
  - While selecting the examples, only such of the salient features of the examples have to be mentioned which are necessary to explain the general characters of the phylum/class.
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## REFERENCES:

1. Adam, S. 1990. A Students Text Book of Zoology, Low Price Publications, Delhi, Vol. I, II & Vol.III.
2. Ayyar, E. 1982. A Manual of Zoology Vol. II, S. Vishwanathan Pvt. Ltd.
3. Colbert, E.H. 2011. Evolution of the Vertebrates, Wiley Student Edition.
4. Dhami & Dhami. 2014. Chordate Zoology, R. Chand & Co. New Delhi.
5. Jordan, E.L. & Verma, P.S. 2013. Chordate Zoology, S. Chand & Company, New Delhi.
6. Kotpal, R.L. 2016. Modern Text Book of Zoology – Vertebrates, Rastogi Publications, Meerut.
7. Lal, S.S. 2009. Practical Zoology Vertebrate 12<sup>th</sup> Ed, Rastogi Publications, Meerut.
8. Parker, T. J. & William, A. H. 1990. A Text Book of Zoology, Low Price Publications, Delhi, Vol. I & II.
9. Prasad, S.N. and Kashyap, V. 2015. A text book of Vertebrate Zoology, New Age International Ltd, New Delhi.
10. Prabhakar Achar, K. & Geetha Nayak, K. 2000. Birds of Dakshina Kannada, Bhuvanendra Nature Club - India.
11. Prakash, M. & Arora, C. K. 1998. Laboratory Animals, Anmol Publicating, Ansari Road, New Delhi.
12. Verma, P.S. 2013. A Manual of Practical Zoology (Vertebrates), S. Chand & Company, New Delhi.

|                                   |                                                   |                                   |              |
|-----------------------------------|---------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                      | <b>SEMESTER</b>                   | <b>II</b>    |
| <b>Course Title</b>               | <b>ANIMAL DIVERSITY-II (CHORDATA) (PRACTICAL)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOPS201</b>                                 | <b>No. of Credits</b>             | <b>2</b>     |
| <b>Contact hours</b>              | <b>4 Hrs/week</b>                                 | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>10</b>                                         | <b>Summative Assessment Marks</b> | <b>40</b>    |

### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Identify and classify important chordate specimens, skeletons, feathers, scales, and museum materials.

**CO2:** Demonstrate understanding of vertebrate anatomy through observation of dissections and organ systems.

**CO3:** Prepare and identify fish scale mountings and interpret their structural differences.

**CO4:** Develop practical skills in comparative anatomy and morphological identification of chordates.

### **Museum specimens and slides**

1. Commonly available specimens cited in the list of examples are to be selected for practicals.
2. Dissections (Demonstration only)

Mouse:

- Digestive system
- Male and Female Urinogenital systems

Fish: Afferent branchial system, Cranial nerves V, VII, X.

3. Mounting: Fish scales (Placoid, Cycloid and Ctenoid)
4. Study of Exoskeletal structures: Bird feathers (Down feather, Contour feather); Horns (Cattle, Sheep); Hooves (Cattle, Horse/ Donkey)
5. Study of Endoskeletal structures: Skull (Amphibian, Reptilian, Avian, Mammalian); Vertebrae, Girdles and limb skeleton of Frog.

### **LIST OF MUSEUM SPECIMENS AND SLIDES**

1. *Balanoglossus*, *Herdmania*, *Amphioxus*, Tornaria larva (all).
2. *Petromyzon*, *Myxine*, *Ammocoetes* larva (all).
3. *Narcine* (Electric ray), *Pristis* (Saw fish), *Trygon* (Sting ray), *Scoliodon* (Shark) (Any two).
4. *Anguilla*, *Hippocampus*, *Anabas testudineus*, *Catla catla*, *Clarius batrachus*, *Gambusia affinis* (Any two).
5. *Ichthyophis* (Caecilian), *Bufo melanosticus* (Common Toad), *Rana hexadactyla* (Indian Pond Frog), *Rana cyanophlyctis* (Skipper Frog), *Hoplobatrachus tigerinus* (Indian Bull Frog), *Rhacophorus malabaricus* (Malabar Gliding Frog), *Ambystoma*, *Salamander*, Axolotl larva (1 limbless; 1 tailed; others -2).

6. *Hemidactylus frenatus* (Southern House Gecko), *Calotes versicolor* (Common garden Lizard), *Varanus benghalensis* (Common Indian Monitor), *Draco dussumieri* (Draco), *Calotes rouxi* (Forest Calotes), *Chameleon zeylancius* (Indian Chaameleon), *Crocodylus porosus* (Mugger) - (any three).
7. *Ptyas mucosus* (Common rat snake), *Bungarus fasciatus* (Banded Krait), *Bungarus caeruleus* (Common Indian Krait), *Naja naja* (Indian Cobra), *Python molurus* (Indian Python), *Trimeresurus malabaricus* (Pit Viper), *Eryx conicus* (Russell's Earth Boa), *Daboia russelii* (Russell's Viper) - (any four).
8. *Geochelone elegans* (Starred Tortoise), *Geochelone travancorica* (Travancore Tortoise), *Chelone mydas* (any two).
9. *Pycnonotus jocosus* (Red whiskered Bulbul), *Dicrurus adsimilis* (Black Drongo or King crow), *Oriolus xanthornus* (Black-hooded Oriole), *Sturnus pagodarum* (Black headed or Brahminy Myna), *Psittacula cyanocephala* (Blossom headed Parakeet), *Haliastur indus* (Brahminy Kite), *Centropus sinensis* (Crow Pheasant), *Ardea cinerea* (Grey Heron), *Corvus splendens* (House Crow), *Passer domesticus* (House Sparrow), *Tyto alba* (Indian Barn Owl), *Ploceus philippinus* (Indian Baya), *Oriolus oriolus* (Indian Golden Oriole), *Pavo cristatus* (Indian Peafowl), *Acridotheres tristis* (Common Myna), *Ardeola grayii* (Indian Pond Heron or Paddy bird), *Nectarinia asiatica* (Indian Purple Sunbird), *Copsychus saularis* (Magpie Robin), *Amaurionis phoenicurus* (Indian Whitebreasted Waterhen), *Dinopium benghalense* (Lesser Golden backed Woodpecker), *Egretta garzetta* (Little Egret) - (any four).
10. *Echidna*, *Bandicota indica* (Bandicoot Rat), *Lepus nigricollis* (Black naped hare), *Macaca radiata* (Bonnet Macaque), *Presbytis entellus* (Common Langur), *Herpestus edwardsi* (Common Mongoose), *Paradoxurus hermaphrodites* (Common Palm Civet), *Petaurista philippensis* (Common giant flying squirrel), *Funambulus palmarum* (Three striped palm squirrel), *Rousettus leshenulti* (Fulvous fruit bat), *Mus musculus* (House Mouse), *Rattus rattus* (House Rat), *Pteropus giganteus* (Indian Flying Fox) - (any three).

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**Note:**

- Local examples with common and scientific names are to be given more emphasis for all the groups.
  - Only such of the salient features of the examples have to be mentioned which are necessary to explain the general characters of the phylum/class.
  - In the event of non-availability of specimens, related internet downloaded photos/movies can be shown.
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**SCHEME OF PRACTICAL EXAMINATION B.Sc. ZOOLOGY: II SEMESTER**

**Course Title: ANIMAL DIVERSITY- II (CHORDATA)**

**Code: BSCZOPS201**

**Duration: 3 hours**

**Max. Marks: 40**

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|                                                                                                                                                                                                                                                                                       |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| I. Dissection - Identify, draw labeled diagram and comment on systems A and B.<br>(Identification of the system - 1 Mark; Labelled diagram of the entire system – 2 Marks; Description - 1 Mark) (A & B in different animals)                                                         | <b>2x4 = 08</b> |
| II. Mounting – Make a stained, temporary mounting of the given material C<br>(Fish Scale)<br>(Stained preparation-2 Marks)                                                                                                                                                            | <b>02</b>       |
| III. Identify, classify, draw labeled diagram and comment on D, E, and F.<br>(Identification-½ Mark; Classification-½ Mark; Labelled diagram-1Mark; Four Comments-2 Marks)                                                                                                            | <b>3x4=12</b>   |
| IV. Exoskeleton – Identify and comment on G.<br>(Identification-1 Mark; Minimum two unique features-1 Mark)                                                                                                                                                                           | <b>02</b>       |
| V. Endoskeleton – Identify and comment on the material H and I<br>(One Skull/Girdle/limb Skeleton; One vertebra).<br>(Skull/Girdle/Limb Skeleton: Identification-1 Mark; Minimum six unique features-3 Marks). (Vertebra: Identification - ½ Mark; Minimum 3 unique features-1½Marks) | <b>4+2=06</b>   |
| VI. Class records                                                                                                                                                                                                                                                                     | <b>10</b>       |
|                                                                                                                                                                                                                                                                                       | <b>TOTAL=40</b> |

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**Note:**

- Questions must be framed as per the scheme provided.
- Internal assessment marks to be allotted after conducting one practical test at the end of the semester.

|                                   |                                                         |                                   |              |
|-----------------------------------|---------------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                            | <b>SEMESTER</b>                   | <b>III</b>   |
| <b>Course Title</b>               | <b>PHYSIOLOGY, BIOCHEMISTRY AND IMMUNOLOGY (THEORY)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOCS301</b>                                       | <b>No. of Credits</b>             | <b>3</b>     |
| <b>Contact hours</b>              | <b>48 Hrs<br/>(4 hours/week)</b>                        | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>20</b>                                               | <b>Summative Assessment Marks</b> | <b>80</b>    |

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### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Explain major physiological processes including osmoregulation, respiration, circulation, excretion, and muscle contraction.

**CO2:** Describe nervous coordination, sensory mechanisms, and endocrine regulation in humans.

**CO3:** Interpret the structure, functions, and biological importance of carbohydrates, proteins, lipids, enzymes, and vitamins.

**CO4:** Analyze the components and functions of the immune system, immunization, and autoimmune disorders.

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### **Unit – I: Physiology**

#### **1.1 Introduction**

**1 Hr**

Definition; Branches and scope of physiology.

#### **1.2 Osmoregulation**

**2 Hrs**

Definition; Types of solutions-hypotonic, isotonic and hypertonic; Osmoconformers and Osmoregulators; Osmoregulation in shark, marine and freshwater teleosts, terrestrial mammals (Kangaroo rat and camel).

#### **1.3 Thermoregulation**

**2 Hrs**

Ectotherms, Endotherms, and Heterotherms; Temperature regulation in Poikilotherms and Homeotherms; Aestivation and hibernation.

#### **1.4 Digestion**

**4 Hrs**

Mechanical and chemical digestion; Digestion and absorption of carbohydrates, proteins and lipids.

#### **1.5 Respiration**

**3 Hrs**

External and internal respiration; Respiratory pigments - Haemoglobin, haemocyanin and haemoerythrin; Physiology of respiration – breathing, exchange of gases - transport of oxygen - oxygen dissociation curves - Bohr effect - transport of carbon dioxide - chloride shift; Respiratory quotient.

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## **Unit - II: Physiology (Contd... )**

### **2.1 Circulation**

**4 Hrs**

Types of circulation; Structure and functions of human heart; Origin and conduction of heart beat; Cardiac cycle; Blood pressure - hypertension and hypotension; Composition of human blood – Plasma, Erythrocytes, leucocytes and platelets.

### **2.2 Nitrogen Excretion**

**4 Hrs**

Nitrogen excretion in aquatic and terrestrial animals – Ammonotelism, Ureotelism and Uricotelism with examples; Ornithine cycle in humans; Physiology of urine formation in humans – ultrafiltration, tubular reabsorption and tubular secretion.

### **2.3 Muscle Contraction**

**4 Hrs**

Principal types of muscles; Ultrastructure of striated muscles; Contractile proteins - myosin, actin, tropomyosin, troponin and actinin; Mechanism of muscle contraction and relaxation - the sliding filament theory; Structure of neuromuscular junction; Properties of muscle - Muscle fatigue, muscle twitch, muscle tetanus, rigor mortis.

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## **Unit - III: Physiology (Contd...)**

### **3.1 Nerve Coordination**

**4 Hrs**

Types of nervous systems; Structure and types of neurons; Nature and conduction of nerve impulse; Types of synapses and synaptic transmission.

### **3.2 Sense Organs**

**4 Hrs**

Classification of sense organs – Photo-, chemo- and thermoreceptors; Structure of mammalian ear and mechanism of hearing; Structure of mammalian eye and mechanism of image formation.

### **3.3 Endocrine System**

**4 Hrs**

Human endocrine glands – Functions of Pituitary, thyroid, parathyroid, pancreas, adrenals, and pineal glands; Hormonal disorders in humans - Pituitary dwarfism, Cretinism, Cushing's disease, Diabetes mellitus; Hypothalamus - stimulating and inhibitory effects.

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## **Unit - IV: Biochemistry and Immunology**

### **4.1 Biochemistry**

**4.1.1 a. Carbohydrates** – Definition, classification, examples; Biological importance.

**b. Lipids** - Definition, classification, examples; Biological importance.

**1 Hrs**

**4.1.2 Proteins** – Definition; A brief account of amino acids; Classification of proteins, examples and biological importance of proteins.

**1 Hrs**

**4.1.3 Enzymes** – Definition, types; Classification of enzymes (IUB system); Mechanism of enzyme action – Lock and key model; Factors affecting enzyme action; Mechanism of enzyme inhibition.

**2 Hrs**

**4.1.4 Vitamins** - Functions of Fat-soluble vitamins (A, D, E and K), water soluble vitamins (B complex and vitamin C); Deficiency symptoms: Night blindness, Scurvy, Beri-beri, Rickets. **2 Hrs**

## **4.2 Immunology**

Immune System - innate and acquired; Cells of immune system; Organs of immune system- Primary lymphoid organs (Bone Marrow, Bursa of Fabricius, Thymus), Secondary lymphoid organs (Lymphoid follicles, lymph nodes, Peyer's patches); Antigens and Antigenicity; Immunoglobulins - structure and functions of IgG; Primary and secondary immune responses; Immunization; Autoimmune diseases – Definition, examples – Type I Diabetes and Rheumatoid Arthritis. **6 Hrs**

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## **REFERENCES:**

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2. Chatterjee, C. C. 2013. Human Physiology, Vol. I., CBS Publishing.
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8. Hoar, W.S. 1983. Comparative Animal Physiology, Prentice Hall.
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11. Kindt, T.J., Goldsby R.A., Osborne, B.A., and Kuby, J. 2006. Immunology, VI Edition, W.H. Freeman Company.
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|                                   |                                                            |                                   |              |
|-----------------------------------|------------------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                               | <b>SEMESTER</b>                   | <b>III</b>   |
| <b>Course Title</b>               | <b>PHYSIOLOGY, BIOCHEMISTRY AND IMMUNOLOGY (PRACTICAL)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOPS301</b>                                          | <b>No. of Credits</b>             | <b>2</b>     |
| <b>Contact hours</b>              | <b>4 Hrs/week</b>                                          | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>10</b>                                                  | <b>Summative Assessment Marks</b> | <b>40</b>    |

### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Perform physiological experiments related to blood analysis, enzyme activity, and osmotic phenomena.

**CO2:** Conduct qualitative biochemical tests for carbohydrates, proteins, nitrogenous waste, and urine constituents.

**CO3:** Identify blood cells and immune organs using stained preparations and specimens.

**CO4:** Develop laboratory skills in handling biological samples, recording observations, and interpreting results.

### **MAJOR EXPERIMENTS:**

#### **A. Physiology:**

1. Total erythrocyte counts in human blood sample.
2. Total leucocyte counts in human blood sample.
3. Salivary amylase activity test of human saliva.
4. Osmotic haemolysis in animal cells.

#### **B. Biochemistry:**

Qualitative tests:

1. Carbohydrates: Molisch's Test, Benedict's test for glucose, Iodine test for starch.
2. Proteins: Biuret test, Xanthophoretic test, Ninhydrin test.
3. Nitrogenous excretory wastes:
  - i. Ammonia - Nessler's reagent test.
  - ii. Urea – Sodium hypobromite test.
  - iii. Uric acid - Folin's uric acid reagent test.
4. Abnormal constituents of Human urine:
  - i. Sugar (glucose) Benedict's test.
  - ii. Albumen – Heller's Nitric acid ring test.
  - iii. Ketone – Rothera's test.

#### **C. Immunology:**

1. Preparation of stained blood smear and identification of different types of blood cells – RBCs, Neutrophils, Lymphocytes, Eosinophils, Monocytes, and Basophils.
2. Identification of organs of immune system – Bone marrow, Thymus, Lymph nodes, Spleen, Peyer's patches, Tonsils – Specimens/slides/charts/models.

\*Note: (Students have to identify the presence of the organic compound in the sample provided, giving the principle of reaction).

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**MINOR EXPERIMENTS:**

1. Preparation of haematin crystals from human blood.
  2. Determination of bleeding time of human blood.
  3. Determination of clotting time of human blood.
  4. Estimation of hemoglobin in human blood (Sahli's method).
  5. Detection of lipids – Solubility test, Greasy spot test.
-

**SCHEME OF PRACTICAL EXAMINATION B.Sc. ZOOLOGY: III SEMESTER**

**Course Title: PHYSIOLOGY, BIOCHEMISTRY AND IMMUNOLOGY**

**Code: BSCZOPS301**

**Duration: 3 hours**

**Max. Marks: 40**

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|                                                                                                                                                                                                                                                                                                                                                                      |                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| I. Physiology experiment (by lots).<br>(Conducting the test - 4 Marks; Principle/Procedure/observation/calculation/Inference – 4 Marks; Result - 2 Marks)                                                                                                                                                                                                            | <b>10</b>         |
| II. Biochemistry experiment (by lots).<br>Conduct suitable qualitative tests for the detection of Organic compounds/ Nitrogenous wastes/abnormal constituents of urine in the sample provided and report (Name of the test - 1 Mark; Principle - 2 Marks; Conducting the test - 3 Marks; Procedure/observation/inference (in tabular form) -3 Marks; Result -1 Mark) | <b>10</b>         |
| III. Immunology: Identify and comment on the Cell A and Organ B. (Common for all)<br>(Cell A: Identification - ½ Mark, Diagram – 1 Mark, Comments – 1 Mark.<br>Organ B: Identification - ½ Mark, Comments – 2 Marks)                                                                                                                                                 | <b>05</b>         |
| IV. Minor experiment (any one) - Common for all.<br>(Experiment - 4 Marks; Report-1 Mark)                                                                                                                                                                                                                                                                            | <b>05</b>         |
| V. Class record                                                                                                                                                                                                                                                                                                                                                      | <b>10</b>         |
|                                                                                                                                                                                                                                                                                                                                                                      | <b>TOTAL = 40</b> |

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**Note:**

1. Questions must be framed as per the scheme provided.
2. Internal assessment marks to be allotted after conducting one practical test at the end of the semester.

|                                   |                                                                 |                                   |              |
|-----------------------------------|-----------------------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                                    | <b>SEMESTER</b>                   | <b>IV</b>    |
| <b>Course Title</b>               | <b>HISTOLOGY, ANIMAL BEHAVIOUR AND APPLIED ZOOLOGY (THEORY)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOCS401</b>                                               | <b>No. of Credits</b>             | <b>3</b>     |
| <b>Contact hours</b>              | <b>48 Hrs<br/>(4 hours/week)</b>                                | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative assessment marks</b> | <b>20</b>                                                       | <b>Summative assessment marks</b> | <b>80</b>    |

### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Describe the histological structure and functions of important mammalian organs.

**CO2:** Explain different types of animal behaviour including migration, social organization, courtship, and parental care.

**CO3:** Analyze the principles and practices involved in dairy, poultry, vermiculture, aquaculture, apiculture, pearl culture, and lac culture.

**CO4:** Evaluate the economic importance and management practices of applied zoological industries.

### **Unit - I: Histology**

**12 Hrs**

Study of histological structure of following mammalian organs.

(a) Stomach (b) Intestine (c) Ovary (d) Testis (e) Liver (f) Pancreas (g) Thyroid (h) Kidney (i) Adrenal (l) Pituitary

### **Unit - II: Animal Behaviour**

#### **2.1 Introduction, Types of Animal Behavior**

**3 Hrs**

Innate behavior - taxes, reflexes, instincts and motivation;

Learnt behavior - habituation, imprinting, conditioned reflexes and insight learning; Biological clock - circadian rhythms.

#### **2.2 Social organization in animals**

**2 Hrs**

Social behavior – Definition; Social behavior in Honey bees, termites, monkeys.

#### **2.3 Animal migration**

**3 Hrs**

Introduction; Advantages of migration; Migration in fishes - anadromous, catadromous; Migration in birds - types of migration, orientation and navigation, preparation for migration.

#### **2.4 Courtship and Nesting Behavior**

**2 Hrs**

Introduction; Courtship behavior in Peacock, River Tern, and Sarus Crane; Types of nests in birds; Nesting behavior in Baya Weaver Bird and Hornbill.

## **2.5 Parental Care**

**2 Hrs**

Parental care in fishes (*Hippocampus*, *Tilapia* species, *Arius* species) and amphibians (*Rhacophorus*, salamander, *Hyla* and *Ichthyophis*).

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## **Unit -III: Applied Zoology**

### **3.1 Dairy**

**4 Hrs**

Introduction; Breeds of dairy animals; Cattle: Milch breeds - Red Sindhi, Gir; Draught breeds - Amrithmahal, Hallikar; Dual purpose breeds - Krishna valley, Ongole; Exotic breeds – HF, Jersey; Buffalo breeds - Nagpuri, Surti; Goat breeds - Jamunapuri, Malabari; Milk - composition and uses; Milk products - Butter, Ghee, Cheese and Paneer - uses; Utility of cattle in agriculture and transport, biogas, fertilizer and gelatin production.

### **3.2 Poultry**

**4 Hrs**

Introduction; Poultry breeds - layers, broilers and dual-purpose breeds with any two examples each; Desi breeds of poultry - Aseel, Chittagong, Kadaknath, Giriraja; Housing management of poultry – intensive and semi-intensive methods; Poultry diseases – Ranikhet, Fowl pox, Tick fever (Spirochaetosis), Fowl cholera; Prevention and control of poultry diseases.

### **3.3 Vermitechnology**

**4 Hrs**

Ecological classification of earthworms (Epigeic, anesic, endogeic); Introduction to vermiculture; Methods of vermiculture – bin and pit methods; Preparation of vermicompost from any organic waste material (weeds, waste, domestic wastes, paper wastes etc.); Uses of vermicompost and vermiwash.

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## **Unit - IV: Applied Zoology (Contd...)**

### **4.1 Aquaculture**

**4 Hrs**

Definition; Techniques of culturing freshwater fishes; Induced breeding and seed fish production; Fish diseases and their control – White spot disease, Gyrodactylosis, Bacterial gill rot; Shrimp culture.

### **4.2 Apiculture**

**4 Hrs**

Introduction; Bee species used for apiculture; Methods of bee keeping –Traditional method, Modern method – Langstroth model; Predators of honey bees; Extraction of honey; Nutritive value and uses of honey; Bee wax and its uses; Diseases of honey bees and control - Nosemosis, American Foulbrood, Thai Sac Brood.

### **4.3 Pearl Culture**

**2 Hrs**

Species of Pearl Oysters and their occurrence; Formation of natural pearl; Pearl producing sites in India; Steps involved in pearl culture – Oyster collection, Rearing, Insertion of nucleus, post-operational care, harvesting.

### **4.4 Lac Culture**

**2 Hrs**

Lac insect – *Kerria lacca* – Structure and life history; Host plants; Cultivation of Lac; Economic importance of Lac.

## REFERENCES:

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|-----------------------------------|--------------------------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                                       | <b>SEMESTER</b>                   | <b>IV</b>    |
| <b>Course Title</b>               | <b>HISTOLOGY, ANIMAL BEHAVIOUR AND APPLIED ZOOLOGY (PRACTICAL)</b> |                                   |              |
| <b>Course Code:</b>               | <b>BSCZOPS401</b>                                                  | <b>No. of Credits</b>             | <b>2</b>     |
| <b>Contact hours</b>              | <b>4 Hrs/week</b>                                                  | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>10</b>                                                          | <b>Summative Assessment Marks</b> | <b>40</b>    |

### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Identify histological structures and prepare permanent slides of mammalian tissues.

**CO2:** Observe and interpret various behavioural patterns and nesting activities in animals.

**CO3:** Identify economically important animal products and applied zoological materials.

**CO4:** Develop field-based observational and reporting skills through visits and practical training.

1. Observation of mammalian histology slides of the following organs: Stomach, Small intestine, Liver, Pancreas, Kidney, Ovary, testis, Thyroid and Adrenal.
2. Preparation of permanent slides of mammalian Stomach, Liver, Pancreas, Small intestine, Kidney, Testis, Ovary, Thyroid and Adrenal of Rat. (Any two slides prepared by the students have to be submitted at the time of examination).
3. Animal behavior
  - a. Social behavior in honey bees – Castes – Queen, Drone, Worker.
  - b. Study of bird's nests – Any four different types of nests.
  - c. Study of Parental care – *Hippocampus*, *Ichthyophis*.
4. Applied Zoology
  - a. Milk products – Curd, Butter, Ghee, Cheese, Paneer.
  - b. Fish products – Fish oil, Fish meal, Fish manure.
  - c. Honey and Bee wax.
  - d. Poultry – Egg, Meat.
  - e. Pearl and Lac.

### **Field oriented practical's:**

1. Study of nesting and roosting places in birds, study of various types of bird nests (as per the availability).
2. Field visit to study animal behavior in natural habitat –
3. Identification of castes/ Study of bee colonies, bee hives/ant colonies, ant nests/ termites and their mounds/ nesting behavior in solitary and social wasps/monkey troops etc.
4. Preparation of vermicompost using different raw materials (such as weeds, paper waste, domestic waste, sugar cane etc.)
5. Visit to honey bee rearing centers.
6. Visit to fish/shrimp breeding centers.
7. Visit to dairy.
8. Visit to poultry farm.

**Note:** The field studies should be based on the above-mentioned topics which shall be allotted at the beginning of semester. Each student shall prepare separate field report which is to be certified by staff in-charge and HOD. It should be submitted during practical examination which shall be evaluated by both internal and external examiners. Field work must be supported by proper documents and photographs of the field visit by individual students.

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**SCHEME OF PRACTICAL EXAMINATION**  
**B.Sc. ZOOLOGY: IV SEMESTER**  
**Course Title: HISTOLOGY, ANIMAL BEHAVIOUR, APPLIED ZOOLOGY**  
**(PRACTICAL)**  
**Code: BSCZOPS401**

**Duration: 3 hours**

**Max. Marks: 40**

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|                                                                                                                                                                    |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| I. Histology - Stain, mount, and identify the paraffin section provided.<br>(Slide preparation - 5 Marks; Identification -1 Mark)                                  | <b>06</b>       |
| II. Histology – Identify, draw labelled diagram and comment on permanent slides A and B.<br>(Identification -1 Mark; Labelled diagram -1 Mark; Comments - 2 Marks) | <b>2x4 = 08</b> |
| III. Animal behaviour: Identify and comment on C and D.<br>(Identification - ½ Mark; Comments - 1½ Marks)                                                          | <b>2x2 = 04</b> |
| IV. Applied Zoology: Identify and comment on E, F and G.<br>(Identification - ½ Mark; Comments - 1½ Marks)                                                         | <b>2x3 = 06</b> |
| V. Field Report*<br>(Introduction -1 Mark; Details of field visited - 1 Mark; Observations including photographs - 3 Marks; References -1 Mark)                    | <b>06</b>       |
| VI. Class Record + Slides                                                                                                                                          | <b>8+2 = 10</b> |
|                                                                                                                                                                    | <b>TOTAL=40</b> |

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\*The Hand written field report (not exceeding 10 A4 size sheets) should include introduction, details of fields visited, detailed account of observations made, original photographs and references.

**Note:**

- Questions must be framed as per the scheme provided.
- Internal assessment marks to be allotted after conducting one practical test at the end of the semester.

|                                   |                                                |                                   |              |
|-----------------------------------|------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                   | <b>SEMESTER</b>                   | <b>V</b>     |
| <b>Course Title</b>               | <b>CELL BIOLOGY AND BIOTECHNOLOGY (THEORY)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOCS501</b>                              | <b>No. of Credits</b>             | <b>3</b>     |
| <b>Contact hours</b>              | <b>40 Hrs<br/>(3 hours/week)</b>               | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>20</b>                                      | <b>Summative Assessment Marks</b> | <b>80</b>    |

### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Explain chromosome organization, nucleic acid structure, DNA replication, transcription, and RNA processing.

**CO2:** Describe plasma membrane organization, cytoskeleton, cell differentiation, and cancer biology.

**CO3:** Analyze the genetic code, protein biosynthesis, mitosis, and meiosis.

**CO4:** Evaluate genetic engineering techniques, biotechnology tools, and their applications in medicine and industry.

### **Unit - I: Cell Biology**

#### **1.1 Introduction**

**1 Hrs**

Definition; Subdivisions of cell biology; Scope of cell biology.

#### **1.2 Chromosomes**

**4 Hrs**

Morphology of chromosomes - centromere, telomere; Heterochromatin and Euchromatin; Types of chromosomes, chromosome number; Ultrastructure of chromosomes - Nucleosome model, Giant chromosome - Polytenic and Lamp brush chromosomes.

#### **1.3 Nucleic Acids**

**5 Hrs**

Introduction - Identification of genetic material - Griffith's experiment, experiments of Avery, MacLeod and McCarty, Hershey-Chase experiment; Chemistry of nucleic acids - structure of DNA - Watson and Crick DNA model; Mechanism of DNA replication - Meselson and Stahl's experiment. DNA repair - Excision repair, photoreactivation and recombinational repairs; Types of RNA; Transcription and RNA processing (Intron splicing and post transcriptional modifications).

### **Unit - II: Cell Biology (Contd...)**

#### **2.1 Plasma membrane and Cell Junctions**

**3Hrs**

Fluid mosaic model of plasma membrane; Functions of plasma membrane; Types of cell junctions; Extracellular matrix.

#### **2.3 Cytoskeleton elements**

**2 Hrs**

Microtubules; Microfilaments; Intermediate filaments - organization and functions.

### **2.3. Cell differentiation**

**1 Hrs**

Definition; Nucleo-cytoplasmic interactions (Experiments with *Acetabularia*).

### **2.4 Cancer and Carcinogenic Agents**

**4 Hrs**

Concept of cancer; Types of cancer; Characteristics of cancerous cells; Oncogenes; Immune system in cancer; Carcinogenic agents - physical, chemical and biological; Strategies of cancer therapy - Immunotherapy, Radiotherapy, Chemotherapy; Role of telomere in cell ageing and cancer; A brief account on apoptosis.

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## **Unit - III: Cell Biology (Contd...)**

### **3.1 Genetic Code and Protein Biosynthesis**

**3 Hrs**

Genetic code: Definition and Properties, Wobble hypothesis; Protein Biosynthesis- Components of protein biosynthesis; Mechanism of protein biosynthesis.

### **3.2. Mitosis**

**3 Hrs**

Cell cycle - Phases of cell cycle – interphase – prophase - centriole cycle – mitotic apparatus - metaphase, anaphase and telophase; Cytokinesis; Differences between mitosis in animal and plant cells; significance of mitosis; Mitotic inhibitors - examples and applications.

### **3.3. Meiosis**

**4 Hrs**

Phases of meiotic cycle - I meiotic division - Synaptonemal complex and recombination; Mechanism of crossing over; cytological basis of crossing over (Stern's experiment); Significance of crossing over; Interkinesis; Second meiotic division; Significance of meiosis; Differences between mitosis and meiosis.

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## **Unit - IV: Biotechnology**

### **4.1 Introduction to Genetic Engineering**

**4Hrs**

Introduction; Restriction endonucleases (I, II and III); Cloning vehicles - plasmids, episomes, transposons, animal viruses, shuttle vectors; Gene libraries -genomic library - cDNA library; Introducing Cloned Genes into the Host Cells - Transformation, Transduction, Particle Gun, Electroporation.

### **4.2 Techniques in Biotechnology**

**3 Hrs**

PCR technique, Southern, Northern and Western blotting technique; DNA finger printing-principle, method and applications; DNA sequencing – Sanger and Cohlson's Method; Proteomics and genomics; Protein engineering; FISH, RAPD, RFLP - Definition and applications.

### **4.3 Applications of Biotechnology**

**3 Hrs**

Introduction; Microbes in pollution control; Biotechnology in bioremediation; Biotechnology in disease prevention and diagnosis, Pharmaceuticals, Vaccines; Transgenic animals; Stem cells in transgenesis; Gene therapy.

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14. Verma and Agarwal. 1999. A Text Book of Cytology, S. Chand & Company. New Delhi.
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| <b>Program Name</b>               | <b>B.Sc.</b>                                                         | <b>SEMESTER</b>                   | <b>V</b>     |
| <b>Course Title</b>               | <b>GENETICS, BIOSTATISTICS, EVOLUTION AND PALAEONTOLOGY (THEORY)</b> |                                   |              |
| <b>Course Code:</b>               | <b>BSCZOC502</b>                                                     | <b>No. of Credits</b>             | <b>3</b>     |
| <b>Contact hours</b>              | <b>40 Hrs<br/>(3 hours/week)</b>                                     | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>20</b>                                                            | <b>Summative Assessment Marks</b> | <b>80</b>    |

### COURSE OUTCOME:

After completion of the course, students will be able to:

**CO1:** Explain Mendelian genetics, gene interactions, linkage, mutation, and sex-linked inheritance.

**CO2:** Describe human genetics, chromosomal abnormalities, pedigree analysis, and genetic counseling.

**CO3:** Apply biostatistical methods for data presentation, analysis, and interpretation.

**CO4:** Analyze theories of evolution, evidence of evolution, fossilization, and evolutionary history of horse and man.

### Unit - I: Genetics

#### 1.1. Introduction and Laws of Inheritance

4Hrs

Introduction and branches of genetics; Heredity and variation; Mendel's experiments - Mono and Dihybrid crosses, Test cross and Back cross; Mendel's laws of inheritance; Use of *Drosophila* in genetic studies; Dihybrid crosses in *Drosophila*.

#### 1.2 Interaction of Genes (Modified genetic ratios)

3Hrs

Incomplete dominance - 1:2:1 - Plumage pigmentation in fowl.

Supplementary factors - 9:3:3:1 - Comb pattern in fowls.

Dominant Epistasis - 13:3 - Plumage colour in Leghorn and Wyandotte.

Recessive Epistasis - 9:3:4 - Coat colour in Guinea pigs.

Complementary factors - 9:7 - Flower colour in sweet peas.

Lethal genes - Coat colour in mice.

#### 1.3 Multiple Alleles, Polygenic inheritance and Pleiotropism

3Hrs

ABO blood groups in humans; Rh factor - Erythroblastosis fetalis; Blood typing and blood transfusion; Inheritance of coat color in rabbit; Polygenic inheritance in man - skin color; Pleiotropism (Cystic fibrosis in humans and vestigial wing in *Drosophila*).

### Unit - II: Genetics (Contd....)

#### 2.1 Nature and Nurture

1 Hrs

Definition; Norm of reaction, Experiments on Himalayan Albino rabbit and Human twins; Phenocopy; Penetrance and expressivity with examples.

## **2.2 Linkage and gene mapping**

**3 Hrs**

Linkage – Definition; Complete and partial linkage in *Drosophila*; Significance of linkage; Linkage maps - Construction of chromosome maps; Two-point test cross, three-point test cross.

## **2.3 Sex-determination and Sex-linked inheritance**

**3 Hrs**

Types of Sex chromosomes; Chromosomal mechanism of sex determination (XX-XY, XX-XO, ZZ-ZW and ZZ-ZO types with specific examples); Sex linked inheritance in *Drosophila* (White eye); Haemophilia and Colour blindness in man; Sex linkage in birds (barred plumage pattern in poultry); Sex limited (plumage pattern in Leghorn fowls) and sex influenced traits (baldness in humans).

## **2.4 Gene, gene regulation and Gene Mutation**

**3Hrs**

Fine structure of gene - cistron, muton, recon, introns; Regulation of gene expression in Prokaryotes – Lac Operon; Mutation - point mutation, frame-shift mutation, insertions and inversions. ClB technique.

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## **Unit - III: Human Genetics and Biostatistics**

### **3.1 Human Genetic**

**4 Hrs**

Human karyotype; Idiogram; Pedigree analysis; Common human chromosomal syndromes- Klinefelter's and Turner's Syndromes, Down's syndrome; Inborn errors of metabolism – Albinism, Phenylketonuria, Alkaptonuria, Sickle cell anemia, Thalassemia, Huntington's chorea; Prenatal diagnosis - Amniocentesis, chorionic villus sampling; Genetic counseling.

### **3.2 Biostatistics**

**6 Hrs**

Introduction to biostatistics - Basic concepts; Presentation of data - Tabulation, frequency distribution, graphical and diagrammatic representation; Analysis of data - mean, median and mode; Standard deviation; Tests of significance - Student t - test, chi-square test. (Wherever necessary, appropriate problems should be worked out).

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## **Unit - IV: Evolution and Palaeontology**

### **4.1 Theories of Organic Evolution and Speciation**

**4Hrs**

Lamarckism; Darwinism; Neo-Darwinism (Factors influencing change in gene frequencies of a population - gene mutation, gene flow, genetic drift); Hardy-Weinberg law of equilibrium; Natural Selection –types - Stabilizing selection, Directional selection and Disruptive selection; Artificial selection with examples. Isolation and Isolating mechanisms – i. Geographical isolation, ii. Reproductive isolation -Prezygotic/Premating isolation – Ecological, Seasonal, Ethological, Mechanical; Post zygotic/Post mating isolation – Cytological, Zygotic mortality, Hybrid inviability, Hybrid sterility; Speciation- sympatric and allopatric speciation.

### **4.2 Evidences of Organic Evolution**

**2 Hrs**

Evidences from comparative morphology and anatomy, comparative physiology and biochemistry, comparative embryology and palaeontology.

### **4.3 Palaeontology**

**2 Hrs**

Brief account of geological time scale; Fossils and fossilization; Dinosaurs; Study of connecting links: *Peripatus* and *Archaeopteryx*.

#### 4.4. Evolution of Horse and Man

2 Hrs

Origin and evolution of horse (*Eohippus*, *Mesohippus*, *Merichyppus* and *Equus*) and man (*Australopithecus*, Java ape man, Neanderthal man and Cro-Magnon Man).

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#### REFERENCES:

1. Dobzhansky, Ayala, Stebbins & Valentine. 1977. Evolution, W.H. Freeman & Company.
2. Gardner. 1991. Principles of Genetics, John Wiley & Sons Inc., New York.
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4. Gurumani, N. 2004. An Introduction to Biostatistics, MJP publishers, Chennai.
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|-----------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                                                                      | <b>SEMESTER</b>                   | <b>V</b>     |
| <b>Course Title</b>               | <b>CELL BIOLOGY, BIOTECHNOLOGY, GENETICS, BIOSTATISTICS, EVOLUTION AND PALAEOLOGY (PRACTICAL)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOPS501</b>                                                                                 | <b>No. of Credits</b>             | <b>2</b>     |
| <b>Contact hours</b>              | <b>4 Hrs/week</b>                                                                                 | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>10</b>                                                                                         | <b>Summative Assessment Marks</b> | <b>40</b>    |

### COURSE OUTCOME:

**After completion of the course, students will be able to:**

**CO1:** Perform cytological techniques such as squash preparations.

**CO2:** Identify stages of mitosis, meiosis, *Drosophila* mutants, and blood groups.

**CO3:** Solve genetics and biostatistics problems using appropriate methods.

**CO4:** Interpret evolutionary and paleontological specimens, models, and adaptations.

1. Observation of permanent slides of onion root tip to study all stages of mitosis.
2. Observation of permanent slides of grasshopper testis to study various stages of meiosis.
3. Squash preparation of onion root tip to demonstrate stages of mitosis.
4. Squash preparation of grasshopper testis to demonstrate stages of meiosis.
5. Squash preparation of salivary gland chromosomes of *Drosophila*/ *Chironomus* larva.
6. Experiments with *Drosophila*.
  - a. Phenotypic characters and sexual dimorphism in *Drosophila*.
  - b. Mounting of sex comb.
  - c. Mutants of *Drosophila* (white eye, bar eye, sepia eye, vestigial wing, curly wing, ebony body and yellow body – any four).
7. Blood typing for the detection of ABO blood group and Rh factor.
8. Genetics problems/ Biostatistics problems
  - a. Dihybrid inheritance (1)
  - b. Multiple alleles - ABO blood group in humans (1)
  - c. Sex linked inheritance in humans (1)
  - d. Graphical/diagrammatic representation (1)
  - e. Mean/median/mode (1)
9. Evolution
  - a. Study of homologous organs - forelimbs of frog and bird;
  - b. Study of analogous organs - wings of insect and bird.
  - c. Study of vestigial organs - appendix and third molar tooth in man. (Any one)
10. Palaeontology
  - a. Connecting links: *Archaeopteryx*.
  - b. Models of Dinosaurs: (Tyrannosaurus, Brontosaurus, Stegosaurus and Triceratops).

## SCHEME OF PRACTICAL EXAMINATION

B.Sc. Zoology: V Semester

Course Title: CELL BIOLOGY, BIOTECHNOLOGY, GENETICS, BIOSTATISTICS,  
EVOLUTION AND PALEONTOLOGY

Code: BSCZOPS501

Duration: 3 hours

Max. Marks: 40

- 
- I. a. Squash - Make a stained squash preparation of onion root tip or grasshopper testis. **05**  
(Stained slide preparation with at least one dividing stage - 4 Marks; Identification - 1 Mark)  
b. Squash - Make a stained squash preparation of salivary gland chromosomes **05**  
(Dissecting the gland – 3 Marks; Slide preparation – 2 Marks)
- II. Identify and comment on the permanent slides A with labeled diagrams. **05**
- III. (Mitosis or Meiosis) (Identification - 1 Mark; Labeled diagram – 2 Marks; Comments -2 Marks)
- IV. Solve the Genetics problem or Biostatistics problem. **05**  
(Working out the problem - 4 Marks; Result – 1 Mark)
- V. Make a temporary mounting of the sex comb. **05**  
(Mounting of entire tarsus with sex comb on a glass slide with cover slip and focused under low power)
- OR**
- Identify the ABO and Rh blood group of the given blood sample and comment on the significance of blood grouping.  
(Identification of ABO and Rh group  $\frac{1}{2} + \frac{1}{2} = 1$  Mark; Reasons  $2+1 = 3$  Marks (student should write the antigen antibody reaction of the identified blood group; Significance - 1 Mark)
- VI. a. Identify the *Drosophila* mutants B with reasons. **02**  
(Identification -  $\frac{1}{2}$  Mark; Chromosome number and site -  $\frac{1}{2}$  Mark; Characters- 1 Marks)  
b. Identify and comment on E (specimen or model from Evolution or Paleontology). **03**  
(Identification 1 Mark; Labeled diagram - 1 Mark; Comments - 1 Marks).
- VII. Class Record **10**
- TOTAL = 40**
- 

### Note:

- Questions must be framed as per the scheme provided.
- Internal assessment marks to be allotted after conducting one practical test at the end of the semester.

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| <b>Program Name</b>               | <b>B.Sc.</b>                                                   | <b>SEMESTER</b>                   | <b>VI</b>    |
| <b>Course Title</b>               | <b>REPRODUCTIVE BIOLOGY AND DEVELOPMENTAL BIOLOGY (THEORY)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOCS601</b>                                              | <b>No. of Credits</b>             | <b>3</b>     |
| <b>Contact hours</b>              | <b>40 Hrs<br/>(3 hours/week)</b>                               | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>20</b>                                                      | <b>Summative Assessment Marks</b> | <b>80</b>    |

### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Explain the structure and functions of reproductive systems, gametogenesis, and reproductive cycles.

**CO2:** Describe fertilization, cleavage, parthenogenesis, and modern reproductive technologies.

**CO3:** Analyze embryological theories, organizer phenomena, and early development of frog and chick.

**CO4:** Interpret placentation, foetal development, hormonal regulation, and extra-embryonic membranes.

### **Unit - I: Reproductive Biology**

#### **1.1 Reproductive system**

**2 Hrs**

Male reproductive system: primary sex organs - male accessory ducts - copulatory organ - accessory glands; Female reproductive system - primary sex organs - female accessory organs - accessory glands - external genitalia; Secondary sexual characters in humans.

#### **1.2 Gametogenesis**

**3 Hrs**

Spermatogenesis - Formation of spermatids - Spermiogenesis - Structure of mature Spermatozoan; Oogenesis - Previtellogenesis and Vitellogenesis - Estrous cycle in non- primate mammals and menstrual cycle in humans; Comparison between spermatogenesis and Oogenesis.

#### **1.3 Parthenogenesis**

**2 Hrs**

Kinds of parthenogenesis - Natural – Arrhenotoky, Thelytoky - automixis and apomixes; Cyclical parthenogenesis in gall wasps and aphids; Larval parthenogenesis in liver flukes; Artificial parthenogenesis; Significance of parthenogenesis.

#### **1.4. Modern trends in Reproduction**

**3 Hrs**

Manipulation of reproduction - Gene bank, Sperm bank, Superovulation, Cryopreservation; *In-vitro* fertilization (IVF) and embryo transfer (ET); Zygote intra fallopian transfer (ZIFT); Intra uterine transfer (IUT); Gamete intra fallopian transfer (GIFT); Intra cytoplasmic sperm injection (ICSI); Intra-uterine insemination (IUI); Artificial insemination (AI); Surrogate mother; Animal cloning; Cloning of Dolly.

## **Unit - II: Developmental Biology**

### **2.1 Introduction**

**3 Hrs**

Definition and scope; Theories of development - Preformation theory, Epigenetic theory, Baer's Law and Biogenetic law; Branches of embryology; Phases of ontogenetic development – Development and differentiation; Patterns of development - Oviparity, ovo- viviparity and viviparity with examples.

### **2.2 Type of Eggs and Sperms**

**1 Hr**

Types of eggs based on amount and distribution of yolk with examples; Mosaic and regulative eggs; Cleidoic egg and its significance; Types of sperms with examples.

### **2.3 Fertilization**

**3 Hrs**

Kinds of fertilization - external, internal, self and cross fertilization with examples; Mechanism of fertilization - approximation of gametes – chemotaxis - fertilizin and antifertilizin – capacitation - acrosome reaction and sperm penetration - activation of ovum- cortical reaction and fertilization membrane formation – Amphimixis; Monospermic & polyspermic fertilization; Significance of fertilization.

### **2.4 Cleavage**

**3 Hrs**

Definition; Types of cleavage - holoblastic and meroblastic; Patterns of cleavage – radial, biradial, spiral and bilateral cleavage with examples; Effects of yolk on cleavage; Types of blastula with examples.

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## **Unit - III: Developmental Biology (Contd...)**

### **3.1 Organizer Phenomenon**

**3 Hrs**

Definition - The amphibian organizer; The organizer's role in development; Potencies of the dorsal lip of the blastopore of amphibian gastrula; Brachet's experiment; Experiment of Spemann and Mangold; Induction of Organizer; Chemical nature of organizer; Parts of organizer; Theories of organizer phenomenon.

### **3.2 Early Development of Frog**

**4 Hrs**

Cleavage - Blastula - Fate maps of Blastula - Gastrulation - Mesogenesis - Notogenesis and Neurulation.

### **3.3 Early Development of Chick**

**3 Hrs**

Structure of hen's egg - cleavage - blastula - gastrulation - origin and structure of primitive streak - structure of 18, 24 and 48 hours chick embryos.

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## **Unit - IV: Developmental Biology (Contd...)**

### **4.1 Extraembryonic Membranes of Chick**

**2 Hrs**

Development, Structure and functions of Yolk-sac, Amnion, Chorion and Allantois.

### **4.2 Placenta**

**3 Hrs**

Definition; Types of placenta- Yolk sac placenta (marsupials), Allantoic placenta - Structure and functions; Morphological and histological classification of placenta with examples.

#### **4.3 Early Development of Human Fetus**

**3 Hrs**

Structure of Graafian follicle; Ovulation; Fertilization; Morula -blastocyst - implantation – gastrulation; Placenta - structure and functions; Twins and multiple births.

#### **4.4 Role of Hormones in Development**

**2 Hrs**

Gonadotropins and their functions; Hormones secreted by testis and ovaries and their functions; Hormones of placenta and their functions.

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#### **REFERENCES:**

1. Adamstone. 1953. Introduction to Vertebrate Embryology, Wiley & Sons.
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| <b>Program Name</b>               | <b>B.Sc.</b>                                                           | <b>SEMESTER</b>                   | <b>VI</b>    |
| <b>Course Title</b>               | <b>ENVIRONMENTAL BIOLOGY, TOXICOLOGY AND WILDLIFE BIOLOGY (THEORY)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOCS602</b>                                                      | <b>No. of Credits</b>             | <b>3</b>     |
| <b>Contact hours</b>              | <b>40 Hrs<br/>(3 hours/week)</b>                                       | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>20</b>                                                              | <b>Summative Assessment Marks</b> | <b>80</b>    |

### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Explain ecological principles including habitats, ecosystems, population, and community ecology.

**CO2:** Analyze environmental pollution, global warming, toxicology, and biomagnification.

**CO3:** Describe zoogeographical distribution, wildlife threats, and conservation strategies.

**CO4:** Evaluate environmental protection laws, biodiversity conservation programmes, and wildlife management practices.

### **Unit - I: Environmental Biology**

#### **1.1 Introduction 2 Hrs**

Definition and subdivisions of ecology; Concept of habitat - Micro-habitat and Macro-habitat; Ecological Niche - Spatial, Trophic and Multidimensional.

#### **1.2 Abiotic Factors 2 Hrs**

Major abiotic factors – Light and Temperature; Adaptation to extreme environment - Effect of light and temperature on animals – Cyclomorphosis. Soil types and profile.

#### **1.3 Biotic Factors 3 Hrs**

Mutualism with examples; Proto co-operation and commensalism with examples; Parasitism - types with examples; Ammensalism and predation - examples and their importance; Competition - intraspecific and interspecific - Gause's principle.

#### **1.4 Habitats 3 Hrs**

Marine habitat - zonation of the sea and ecological classification of marine biota; Coastal ecology; Estuarine ecology and mangroves; Freshwater habitat - lentic and lotic systems; Ecological classification of freshwater animals; Terrestrial habitats - A brief account of biomes.

### **Unit - II: Environmental Biology (Contd...)**

#### **2.1 Population Ecology 2 Hrs**

Population attributes - population density - natality and mortality; Age distribution, age pyramids; Population growth rate - population growth curves - logistic and exponential; Biotic Potential - Allee's principle.

## **2.2 Community Ecology**

**3 Hrs**

Community structure; Ecological determinants; Ecotone and edge effect; Ecological stratification; Alpha, beta, and gamma diversity; Shannon Index and Simpson's Index; Significance of biodiversity indices.

## **2.3 Ecosystem**

**5 Hrs**

Types of ecosystems with examples; Natural ecosystems; Man engineered ecosystems; Microecosystem; Biosphere and ecotone; Ecosystem – Structural components; Functions of ecosystem – productivity – primary and secondary, decomposition, energy flow – I and II laws of thermodynamics; Food chains – types with examples; Food webs with examples; Ecological pyramids – Types with examples; Nutrient cycling – nitrogen, carbon and phosphorus.

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## **Unit – III: Environmental Pollution and Toxicology**

### **3.1 Environmental Pollution with Reference to India**

**4 Hrs**

Air pollution – Major air pollutants (Carbon dioxide, oxides of sulphur and nitrogen); Control of gaseous pollutants – combustion – absorption – adsorption; Control of particulate pollutants - filters, electrostatic precipitators, cyclone separators, scrubbers and catalytic converters; Air (Prevention and Control of Pollution) Act; Auto fuel policies in India - Bharath stage; A brief account of Acid rain, photochemical smog and ozone hole; Montreal protocol; Control of water pollution - Sewage/ effluent treatment – primary and secondary treatments; Water Act – River action plan; A brief account on BOD and Eutrophication; Solid waste management; Bioindicators and geindicators - definition and example; Environmental Protection Act 1986.

### **3.2 Global Impacts**

**2 Hrs**

Climate change- Global warming- Sources, effect and control measures-Kyoto protocol.

### **3.3 Toxicology**

**4 Hrs**

Definition; Major subdivisions of toxicology and dose response curve; Toxicological parameters - acute and chronic toxicity; LD50, LC50; Factors influencing toxicity - route of administration, host factors-species, age, sex; Bioactivation and detoxification of xenobiotics - types of xenobiotics; Mechanism of biotransformation - phase I and II reactions; Pesticide toxicity - toxicity of organophosphate, organochloride, carbamate and pyrethroid pesticides citing two examples for each; Biomagnification - Biomagnification of DDT and Mercury; Antidotal therapy - Definition and types of antidotes with examples.

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## **Unit - IV: Wildlife Biology**

### **4.1. Zoogeography and Distribution of Wildlife**

**3 Hrs**

Zoogeographical realms of world with climatic conditions and examples of characteristic fauna; A brief account of Wallace's line; Continuous and discontinuous distributions with examples; Barriers of dispersal - topographic and vegetation barriers - large bodies of water as barriers – climatic barriers.

### **4.2 Threats to Wildlife**

**1 Hrs**

Anthropogenic factors - Hunting, overharvesting, habitat destruction, degradation, habitat shrinkage, climate change; Human animal conflict.

### 4.3 Wildlife Conservation

6 Hrs

Agencies engaged in wildlife conservation - Government organizations and non-government organizations (NGOs); Wildlife (protection) Act 1972; CITES (Convention on International Trade in Endangered Species of Wildlife Flora and Fauna); Endangered fauna of India; IUCN categories of endangered animals; Unique Indian animals; Endemic species with examples; Red Data Book; Ramsar convention; CBD; Biosphere reserves - Important National Parks and Wildlife sanctuaries of India (with special emphasis on Karnataka); Special Projects - Project Tiger; Project Elephant; Project Rhino; Biodiversity Protection Act.

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2. Arora, M.P. 2001. Ecology, Himalaya Publishing House, New Delhi.
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6. Habermehl, G.G. 1981. Venomous Animals and Their Toxins, Springer-Verlag, Berlin.
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11. Prater, S.H.1971. The Book of Indian Animals, BNHS, Oxford University Press.
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13. Sharma, P. D. 1999. Toxicology. Rastogi Publishers, Meerut.
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|                                   |                                                                                                                               |                                   |              |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                                                                                                  | <b>SEMESTER</b>                   | <b>VI</b>    |
| <b>Course Title</b>               | <b>REPRODUCTIVE BIOLOGY, DEVELOPMENTAL BIOLOGY<br/>ENVIRONMENTAL BIOLOGY, TOXICOLOGY AND<br/>WILDLIFE BIOLOGY (PRACTICAL)</b> |                                   |              |
| <b>Course Code</b>                | <b>BSCZOPS601</b>                                                                                                             | <b>No. of Credits</b>             | <b>2</b>     |
| <b>Contact hours</b>              | <b>4 Hrs/week</b>                                                                                                             | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b> |
| <b>Formative Assessment Marks</b> | <b>10</b>                                                                                                                     | <b>Summative Assessment Marks</b> | <b>40</b>    |

### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Identify developmental stages, types of eggs, sperms, and placental structures.

**CO2:** Perform ecological and environmental experiments including estimation of dissolved gases and water quality parameters.

**CO3:** Interpret ecological adaptations, biotic relationships.

**CO4:** Develop laboratory and field skills in developmental biology, ecology, and environmental studies.

- Study of different types of eggs and sperms:
  - Type of eggs: Insect, Amphioxus, Frog, Chick
  - Type of sperms: Frog, Domestic fowl, Rat.
- Stages of development of frog: Study of cleavage stages, blastula, gastrula and neurula (sections)
- Study of permanent slides of chick embryo: 18 hrs, 24 hrs, 36 hrs and 48 hrs (WM) chick embryo.
- Study of permanent slides/charts of histological types of placenta (All five)
- Study of charts or models of morphological types of placenta: Diffuse, cotyledonary, intermediate, zonary and discoidal placenta.
- Water quality parameters assessment:
  - Estimation of Dissolved Oxygen (O<sub>2</sub>)
  - Estimation of Carbon dioxide (CO<sub>2</sub>)
  - Estimation of hardness of water.
- Study of ecological adaptations and morphological peculiarities: Hermit crab, Stick insect, Puffer fish, Exocoetus, Phrynosoma, Draco.
- Study of biotic relationships: Leguminous plants, Termites, Liver fluke, Tape worm, Sucker fish, Insectivores plant.
- Identification of wild animals: Pugmarks (Male and Female tigers) and hoof marks (gaur); Scates (elephants) and pellet counts (deer); Antlers (sambhar). (Photographs)
- Study of threatened animals of India (by models/pictures/charts): Sarus crane, Great Indian bustard, Nilgiri tahr, Gharial, Asiatic lion, Indian pangolin. (With reference to their reasons for depletion, importance, significance, and conservation measures). (Photographs)

**SCHEME OF PRACTICAL EXAMINATION B.Sc. ZOOLOGY: VI SEMESTER**  
**Course Title: REPRODUCTIVE BIOLOGY, DEVELOPMENTAL BIOLOGY,**  
**ENVIRONMENTAL BIOLOGY, TOXICOLOGY AND WILDLIFE BIOLOGY**  
**Code: BSCZOPS601**

**Duration: 3 hours**

**Max. Marks: 40**

|                                                                                                                                    |                   |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| I. a. Identify, draw labeled diagram and comment on permanent slides of developmental stages A and B. (1 from frog + 1 from chick) | <b>2 x 4 =08</b>  |
| (Identification -1 Mark, Labeled diagram -1 Marks, Comments - 2 Marks)                                                             |                   |
| b. Identify, draw labeled diagram and comment on C (Egg or Sperm)                                                                  | <b>03</b>         |
| (Identification - 1 Mark, Labeled diagram -1 Mark, Comments - 1 Marks)                                                             |                   |
| II. Identify, draw labelled diagram and comment on the given placental charts/slides/models/specimens D.                           | <b>04</b>         |
| (Histological/Morphological placenta) (Identification -1 Mark, Labeled diagram – 1 Mark, Comments – 2 Marks)                       |                   |
| III. Estimation of dissolved O <sub>2</sub> /CO <sub>2</sub> /Hardness of water.                                                   | <b>06</b>         |
| (Principle-1 Mark, Procedure-2 Marks, Readings & calculation-2 Marks, Results-1 Mark)                                              |                   |
| IV. a. Identify and comment on the ecological adaptations and morphological peculiarities of E.                                    | <b>03</b>         |
| (Identification- 1 Mark, Comments-2 Marks)                                                                                         |                   |
| b. Comment on the biotic relationships of F.                                                                                       | <b>03</b>         |
| (Identification- 1 Mark, Comments-2 Marks)                                                                                         |                   |
| V. Identify and comment on G.                                                                                                      | <b>03</b>         |
| (Question from Expt.-9/10) (Identification- 1 Mark, Comments-2 Marks))                                                             |                   |
| VI. Class Record                                                                                                                   | <b>10</b>         |
|                                                                                                                                    | <b>TOTAL = 40</b> |

**Note:**

- Questions must be framed as per the scheme provided.
- Internal assessment marks to be allotted after conducting one practical test at the end of the semester.

|                                   |                                                                       |                                   |                |
|-----------------------------------|-----------------------------------------------------------------------|-----------------------------------|----------------|
| <b>Program Name</b>               | <b>B.Sc.</b>                                                          | <b>SEMESTER</b>                   | <b>IV/V/VI</b> |
| <b>Compulsory Course</b>          | <b>SKILL ENHANCEMENT PROGRAMME-<br/>PROJECT WORK / FIELD TRAINING</b> |                                   |                |
| <b>Course Code</b>                | <b>BSCZOPRS<br/>401/501/601</b>                                       | <b>No. of Credits</b>             | <b>2</b>       |
| <b>Contact hours</b>              | <b>2 Hrs/week</b>                                                     | <b>Duration of SEA/Exam</b>       | <b>3 Hrs</b>   |
| <b>Formative Assessment Marks</b> | <b>10</b>                                                             | <b>Summative Assessment Marks</b> | <b>40</b>      |

### **COURSE OUTCOME:**

After completion of the course, students will be able to:

**CO1:** Plan and conduct field-based or laboratory-based zoological investigations.

**CO2:** Collect, analyze, and interpret biological data using scientific methods.

**CO3:** Prepare scientific reports/dissertations with proper documentation, references, and photographs.

**CO4:** Develop communication, presentation, and research skills through project presentation and viva voce.

### **Proposed topics for project work/field training:**

1. Study of animal diversity in various habitats - Gardens/croplands/grasslands/forests/ponds/rivers/streams/seashores.
2. Bird watching and preparation of checklist of birds from different habitats.
3. Identification and study of local edible fishes.
4. Listing and identifying local butterflies and preparation of checklist of butterflies.
5. Listing and identifying common spiders and ants.
6. Listing and identifying terrestrial and freshwater mollusks.
7. Identification of molluscan shells from nearby coasts.
8. Diversity of ornamental fishes.
9. Diversity of insects.
10. Insect pests of vegetables, fruit crops, horticultural plants, paddy etc.
11. Store pests.
12. Study of biodiversity in sacred groves.
13. Study of community: By quadrat method to determine frequency, density and abundance of different species present in the community.
14. Diversity of mosquito species.
15. Diversity of wild varieties of *Drosophila*.
16. Listing and studying the use of fertilizers and pesticides in agricultural fields.
17. Identification and behavioral study of wasps.
18. Identification and study of local edible shell fishes.
19. Study of simple Mendelian traits in human populations.
20. Studying organic farming.
21. Study of soil fauna.
22. Evaluation of larvicidal / adulticidal properties of chemicals or plant extracts on mosquitoes or other pests.

**Note:** In addition to the above-mentioned exercises any faunal diversity of local interest and their characteristic features/Environmental pollution problems/Toxicological problems/Human population studies/Ecosystem studies/Utilization of energy resources/Wildlife depletion and conservation topics may be chosen for project work.

**Note:** During field studies care should be taken not to disturb/remove the specimens/nests etc.

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### **Guidelines for project work**

1. Project allotment should be done at the beginning of semester and topic should be finalized in consultation with the guide by the student.
2. Each project will be carried out as individual (preferably) or in a batch of 2/3/4 students. There shall not be more than 4 students in each group.
3. Dissertation work must be submitted in the prescribed format.  
Title page, Page I - Certificate, Page II - Declaration, Page III - Acknowledgements, Page IV – contents followed by the body of the dissertation.
4. Contents should include the following subheadings:  
Introduction with Review of Literature  
Materials and Methods  
Result and Discussion.  
Summary  
References  
Plates containing original photographs (Minimum 6 photos/page)  
Annexure (not compulsory)
5. Dissertation should contain a minimum of 20 pages excluding photographs (A4 sheets with 1 inch margin on all sides, Times New Roman font, font size -12 and line spacing - 1.5).
6. A student who is going to other institutions/industry/laboratory/fields for any assistance has to take permission letter by the HOD/Principal of the college.
7. Dissertation has to be submitted individually even if the work is done in group, i.e. one student has to submit his/her dissertation exclusively. No joint author submission. The dissertation to be certified by project guide and HOD. Certified dissertation shall be submitted during practical examination which shall be evaluated by both internal and external examiners.

**SCHEME OF EXAMINATION B.Sc. ZOOLOGY: VI SEMESTER  
COMPULSORY PAPER - SKILL ENHANCEMENT (PROJECT WORK/FIELD  
TRAINING)**

**Code: BSCZOPRS401/501/601**

**Duration: 3 hours**

**Max. Marks: 40**

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- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| I. Field work and preparation of dissertation<br>(To be evaluated by project guide* + internal examiner** + external examiner** each for 20 marks and average shall be taken).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>20 Marks</b> |
| II. Project guide shall assess the candidate based on his/her involvement in the field work and preparation of dissertation. Marks allotment for the same shall be handed over to HOD in a sealed cover which will be transmitted to examiners.<br>** Distribution of marks: Introduction with review of literature - 3 Marks, Materials and methods - 3 Marks, Result and discussion - 8 Marks, Summary – 1 Mark, References – 1 Mark, Plates containing original photographs - 4 Marks)<br>Presentation by the candidate* (PPT slides preferably or charts)<br>(Preparation of PPT slides or charts – 4 Marks, presentation – 6 Marks)<br>*Presentation to be done individually even if the project is carried out in team. |                 |
| III. Viva-voce (Based on the contents of dissertations)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>10 Marks</b> |
| <b>TOTAL = 40 Marks</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |
- 

**Note:**

- Internal assessment marks to be allotted based on the preparation and presentation of the dissertation topic.

**SCHEME OF EXAMINATION: B.Sc. - I to VI SEMESTERS (THEORY) CORE  
SUBJECT: ZOOLOGY**

| <b>Question No.</b> | <b>PART - A</b>                                                                       | <b>Marks</b> |
|---------------------|---------------------------------------------------------------------------------------|--------------|
| I                   | Answer any TEN Questions out of TWELVE Questions<br>(Give 3 questions from each unit) | 10 x 2 = 20  |
|                     | <b>PART - B</b>                                                                       |              |
|                     | <b>Unit - I</b>                                                                       |              |
| II                  | 4 Marks Questions (Answer any TWO out of THREE)                                       | 2 x 4 = 8    |
| III                 | 7 Marks Questions (Answer any ONE out of TWO)                                         | 1 x 7 = 7    |
|                     | <b>Unit - II</b>                                                                      |              |
| IV                  | 4 Marks Questions (Answer any TWO out of THREE)                                       | 2 x 4 = 8    |
| V                   | 7 Marks Questions (Answer any ONE out of TWO)                                         | 1 x 7 = 7    |
|                     | <b>Unit - III</b>                                                                     |              |
| VI                  | 4 Marks Questions (Answer any TWO out of THREE)                                       | 2 x 4 = 8    |
| VII                 | 7 Marks Questions (Answer any ONE out of TWO)                                         | 1 x 7 = 7    |
|                     | <b>Unit - IV</b>                                                                      |              |
| VIII                | 4 Marks Questions (Answer any TWO out of THREE)                                       | 2 x 4 = 8    |
| IX                  | 7 Marks Questions (Answer any ONE out of TWO)                                         | 1 x 7 = 7    |

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**GROUP II: ELECTIVE (SUPPORTIVE TO THE DISCIPLINE OF STUDY)**  
**BSCZOES301: PARASITOLOGY AND VECTOR BIOLOGY**  
**(To be studied in the III Semester B.Sc.)**  
**(Hours of instruction: 2 hours per week. Total: 24 Hours)**  
**Syllabus**

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**COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Describe the causative organisms, transmission, pathogenicity, and control measures of common parasitic diseases.

**CO2:** Explain the biology and importance of vectors in disease transmission.

**CO3:** Analyze vector-borne diseases and integrated vector management strategies.

**CO4:** Identify arthropod, avian, and mammalian vectors and evaluate methods for their control.

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**Unit - I: Parasitology:**

**12 Hrs.**

Brief account of the disease caused, mode of infection, transmission, pathogenicity and control measures of following parasites:

Protozoan: *Giardia* and *Trichomonas*.

Helminthes: *Taenia* and *Ancylostoma*.

Bacterial: Typhoid and Cholera.

Viral: Hepatitis and H1N1.

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**Unit-II: Vector Biology and Integrated Vector Management**

**12Hrs.**

Vectors: Introduction, types of vectors with examples; Tools for vector control, Sources – biological, chemical, adulticides and larvicides.

Causative organism, transmission and control measures of following vector borne diseases: Filariasis, Japanese Encephalitis, Dengue and Chikungunia.

Arthropods vectors: Diseases transmitted and control Mosquitoes with reference to Mosquitoes (*Anopheles*, *Culex*, *Aedes* and *Mansonia*), Sand fly, Fleas, Stable or Dog Fly, Tsetse Fly, Bed Bugs, Cockroach, House Fly, Human Louse, Bed Bug, Ticks and Mites.

Avian and mammalian vectors: diseases transmitted and control measures with reference to Domestic Fowl, Rat, Bat, Dog, Cattle, Monkey.

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**REFERENCES:**

1. Apurba, S. S and Sandhya, B. 2014. Essentials of Medical Parasitology, Jaypee Brothers Medical Publishers.
2. Arora, D.R. and Arora, B. 2001. Medical Parasitology, CBS Publications.
3. Chandler, A.C. & Read, C.P. 1961. Introduction to Parasitology, John Wiley & Sons Inc.
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6. David, D.V. and Kumara Swami. 1988. Elements of Economic Entomology, Popular Book Depot, Madras.

7. Mathews, G. 2011. Integrated Vector Management: Controlling Vectors of Malaria and Other Insect Vector Borne Diseases, Wiley-Blackwell.
8. Pedigo, L.P. 2002. Entomology and Pest Management, Prentice Hall Publication.
9. Smyth, J.D. 1994. Introduction to Animal Parasitology, Cambridge University Press.
10. William, H. M., Richard, S. D., Robert, B. G. 1999. Parasitology & Vector Biology, Academic Press.



## **GROUP II: ELECTIVE (NURTURING STUDENT'S PROFICIENCY/SKILL)**

### **BSCZOES401: AQUARIUM FISH KEEPING**

**(To be studied in the IV Semester B.Sc.)**

**(Hours of instruction: 2 hours per week. Total: 24 hours) Syllabus**

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#### **COURSE OUTCOME:**

**After completion of the course, students will be able to:**

**CO1:** Explain the biology, diversity, and sexual dimorphism of ornamental aquarium fishes.

**CO2:** Describe the construction, setup, and maintenance of aquaria and water quality management.

**CO3:** Analyze feeding practices, reproductive biology, and fish seed collection techniques in aquarium fishes.

**CO4:** Evaluate fish transportation methods and management of aquarium fish diseases.

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#### **Unit - I: Introduction to Aquarium Fish Keeping**

##### **Biology of Aquarium Fishes**

**6 Hrs**

Importance and scope of aquarium fish keeping; Exotic and endemic species of aquarium fishes; Common characters and sexual dimorphism in aquarium fishes: Guppy, Molly, Sword tail, Goldfish, Angel fish, Koi, Gourami, Zebra fish, Fighter fish.

Aquarium Setup      6 Hrs

Construction and preparation – size, shape, substrate, ornamental aquatic plants, bio- filters, aerators; Accessories for fish tank - hood and light, nets, suction tube, feeding cups and breeding traps; Water quality management - pH, hardness, salinity, oxygen, carbon dioxide, chlorine, ammonia and temperature.

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#### **Unit - II: Aquarium Management**

##### **Food, Feeding and Reproduction**

**6 Hrs**

Nutritional requirement of fishes, fish feed, composition of fish feed; Feeding methods- live feed, artificial feed; Reproductive Biology of Goldfish, Fish seed collection.

Transportation and Disease Management      6 Hrs Live fish transport - fish handling, packing and forwarding techniques; Aquarium fish diseases - Bacterial, Viral, Fungal and Protozoan infections, treatment and control.

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#### **REFERENCES:**

1. Ahilan, B., Felix, N., Santhnam R. 2008. Textbook of Aquaculture, Daya Publishing House.
2. Alappat, H. J. & Biju, K. 2011. Aquarium Fishes: A Colourful Profile, B.R. Publishing Corporation.
3. Amita, S. 2003. Aquarium Management, Daya Publishing House.
4. Biju, K. & Alappat, M. 1995. A Complete Guide to Aquarium Keeping, Low Price Publications.
5. David, A. 2011. Encyclopedia of Aquarium & Pond Fish, Penguin, UK.
6. Rolf, G. 1963. Aquarium Fish Diseases, TFH Publications.
7. Tharadevi, C.S., Jayashree, K.V., Arumugam, N. 2015. Home Aquarium and Ornamental Fish Culture, Saras Publication.
8. Ulrich, S. 2005. Tropical Freshwater Aquarium Fish from A to Z, Barron's Educational Series Inc., U.S.
9. Uma., Felix & Gopalakannan. 2018. Fish Diseases & Management, Tamil Nadu Dr. Jayalalitha University.

**SCHEME OF EXAMINATION  
ELECTIVE PAPERS (BSCZOES301 and BSCZOES401)**

| <b>Question No.</b> | <b>PART - A</b>                                                                     | <b>Marks</b> |
|---------------------|-------------------------------------------------------------------------------------|--------------|
| I                   | Answer any FIVE Questions out of SIX Questions<br>(Give 3 questions from each unit) | 5 x 2 = 10   |
|                     | <b>PART - B</b>                                                                     |              |
|                     | <b>Unit - I</b>                                                                     |              |
| II                  | 4 Marks Questions (Answer any TWO out of THREE)                                     | 2 x 4 = 8    |
| III                 | 7 Marks Questions (Answer any ONE out of TWO)                                       | 1 x 7 = 7    |
|                     | <b>Unit - II</b>                                                                    |              |
| IV                  | 4 Marks Questions (Answer any TWO out of THREE)                                     | 2 x 4 = 8    |
| V                   | 7 Marks Questions (Answer any ONE out of TWO)                                       | 1 x 7 = 7    |

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